

Analyzing and Predicting the Outcomes of UFC Fights: A Statistical Approach

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1 Introduction

Since the inception of the Ultimate Fighting Championship (UFC) in 1993, it has been the home of the greatest combat sports athletes around the globe. The UFC would singlehandedly help popularize the newfound bloodsport at the time known as mixed martial arts (MMA), allowing martial artists of all disciplines to compete against one another in a no-holds-barred fight. Its original goal was simple: to determine which martial arts system is the best and who is the best fighter in the world.

However, even with the advent of the UFC, the answers to these questions are not clear due to the internal politics of the UFC organization. Fighter rankings are determined by non-quantifiable metrics behind closed doors by combat sports journalists. Matchmaking by the UFC has also been heavily criticized by fans as the UFC top brass is seemingly more interested in making fights between who they think will sell the most tickets rather than who is most deserving.

In this report, we will seek to cut through the meddling and chicanery of the UFC and provide the answers to questions that have furious sparked debate in the MMA community for years using a statistical lens:

1. Who are the current top UFC fighters?
2. How has the skill level of UFC fighters changed over time?
3. What stylistic factors improve a fighter's chance of winning?
4. Who would win in a hypothetical match between two UFC fighters?

2 Data Scraping

Instead of using pre-scraped datasets which often don't include the entire repertoire of UFC fights, we decided to scrape the data ourselves. All data we scraped

come from the official UFC Stats website [4]. We divided this task into two separate parts: scraping the statistics for each fight and scraping data on each UFC fighter.

2.1 UFC Fight Statistics

The UFC provides relatively detailed round-by-round statistics for every fight. However, these fight stats are usually tabulated by UFC officials who watch the fight ringside. This means that the data collection may not be completely precise since the officials may not have the best view of the fight or miscalculate certain stats by human error. Nevertheless, we will assume the statistics will be accurate enough for our use cases.

Before we scraped the fight data, we had to collect the URLs for each fight first which was tricky as we had to comb through the approximately 700 UFC events to do so. Once that was done, it became a trivial exercise to scrape all of the statistics we wished to keep. Some examples of these include who won the fight and by what method, how many rounds the fight lasted, and the number of significant strikes landed by the red fighter or the amount of grappling control time by the blue fighter in a specific round. This amounted to 229 features for each fight. In the end, we scraped the data for every single recorded UFC fight as of April 5, 2025, totaling 8126 UFC fights.

2.2 UFC Fighter Data

Scraping the data for each fighter was much easier as the UFC sorts them alphabetically, meaning we only had to go through 26 pages. There were only eight features we were interested in collecting which included name, height, and reach. All in all, we scraped the data for all 4301 fighters who were signed at some point by the UFC.

3 Data Cleaning

Before we could generate features using our scraped data, we had to sanitize, remove nonsensical values, and convert others into a form that could be used by our predictor later on. Once again, we divided this task into two parts, one for the fight data and the other for the individual fighter statistics.

3.1 Cleaning Fight Data

The data from UFC fights was relatively easy to clean. The procedure was as follows:

- Empty entries were set to 0. Examples of these entries would be round statistics for rounds that did not occur in a fight as they ended early.

- Fight results for the red fighter were converted to be either 1, 0.5, or 0, representing a win, loss, or draw respectively for our Glicko rating generator.
- Parsed fight formats to find how many rounds and how long they are for each fight to calculate the total elapsed time.
- Added nicknames to fighters who shared the same name. This only applied to two UFC fighters so we fixed this manually for simplicity; in hindsight, we could have scraped their official nicknames from their fight pages and added them in for robustness.
- Removed fights that ended in by no contest as these typically happen when a fighter gets cut by accident or a fighter fails a drug test. Stats from these fights shouldn't impact a fighter's career so they were deleted.
- Dropped columns that wouldn't be used for feature generation to save space.

This process slimmed down the fight data file to 8040 fights and 226 features.

3.2 Cleaning Fighter Data

Sanitizing the fighter data was also a straightforward process. The following steps were applied:

- The height and reach of a fighter are converted from a formatted string to inches with integer representation. If either is missing, reach is set equal to the height and vice versa. If both were missing, the average of the fighter's weight class is taken. If the weight class was also missing, the fighter is assumed to be in the mode weight class.
- Fighter stances are encoded with integers while missing stances are assumed to be orthodox, the most popular stance.
- Fighters with the same name are handled on a special case-by-case basis by including the nickname of the less well-known fighter (e.g. Jean 'White Bear' Silva). This too was done manually as there was only a handful of fighters who shared the same name. In the future, the official nickname of fighters should be scraped as well to prevent this situation from occurring.
- Columns such as weight and record are dropped as they are not used later on.

4 Generating Features

To reduce the size and complexity of the fight data, we needed to summarize our statistics before training a model with it.

4.1 Summarizing Round-by-Round Statistics

Round-by-round stats are first calculated by adding raw per-round fight data. All relevant stats, such as knockdowns (kd), total strikes thrown, takedowns, control time, etc., are tallied separately for every round for both fighters, i.e., `r1_red_kd`, `r2_red_kd`, ..., `r1_blue_kd`, `r2_blue_kd`, etc. Individual round stats are then tallied across all rounds to generate fight totals for both blue and red fighters. This process of summation ensures that a complete picture of a fighter’s performance in a fight is recorded in detail. Once these totals are calculated, we take the per-round averages by dividing each total stat by the number of rounds fought.

In addition to raw and average counts which capture the magnitude of a fighter’s performance, we calculate the efficiency of stats such as strikes and takedowns landed as percentages, capturing the frequency, by dividing the total number of strikes or takedowns landed by the number of attempts. As a result, we reduced the number of features in our fight data to 61.

4.2 Computing Career Statistics

After all fight-level stats are calculated, career averages are computed by aggregating data into fighter names and averaging every statistic across all their fights. To counterbalance variability issues from fighters with very few official UFC bouts, a Bayesian smoothing algorithm is applied to the average. This method averages each fighter’s observed average with the world mean, weighted by the number of fights they have had. This keeps low data volume fighters from producing misleading outlier-like averages. Ultimately, the process produces a complete and statistically valid summary of each fighter’s career performance which can be used to predict hypothetical fights between another fighter as we will discuss in §6.

4.3 Rating Fighters using Glicko-2

Glicko-2 [1] is an updated version of the famous Elo rating system used to rank chess players, offering improvements in the reliability of its ratings. It consists of three parameters, rating r , rating deviation RD , and volatility σ , initialized to 1500, 350, and 0.06 respectively. The true strength of a competitor has a 95% chance of being within the confidence interval $[r \pm 2 \cdot RD]$.

Our feature generation program implements a modified version of Glicko-2, omitting the concept of rating periods. This decision was made because UFC fighters do not fight often enough—fighters typically fight less than twice a year on average—to make use of a rating period. Instead, a fighter’s Glicko values are updated after every fight and decay occurs if a fighter hasn’t fought within a year.

We also calculated the predicted probability that the red fighter would beat the blue fighter for every UFC fight using their Glicko ratings [2] since we believed it would be a good feature for our predictor.

5 Analyzing Data

5.1 Ranking UFC Fighters by Glicko Rating

As mentioned in §1, the official UFC rankings are determined by the eye test—how good a fighter looks in their fights—rather than using any quantifiable metrics. This routinely results in the UFCs biggest stars being pushed onto the list, effectively stealing a spot from more deserving contenders, potentially damaging their career earnings, and preventing them from challenging for championship titles. We seek to remedy this problem by ranking fighters using Glicko ratings which will hopefully provide an objective pecking order of the current top UFC fighters.

Rank	UFC	Glicko	Rating	RD
1	Islam Makhachev	Jon Jones	2312.869	91.83003
2	Jon Jones	Ilia Topuria	2287.175	142.2054
3	Ilia Topuria	Shavkat Rakhmonov	2251.347	146.6009
4	Merab Dvalishvili	Dricus Du Plessis	2231.284	121.1501
5	Dricus Du Plessis	Islam Makhachev	2230.431	102.11
6	Magomed Ankalaev	Alex Pereira	2194.638	122.5383
7	Belal Muhammad	Ciryl Gane	2159.662	116.2971
8	Alex Pereira	Movsar Evloev	2157.214	133.1581
9	Alexandre Pantoja	Khamzat Chimaev	2156.352	132.6693
10	Tom Aspinall	Stipe Miocic	2146.229	94.72055

Table 1: Top 10 UFC P4P fighters [3] compared to the top 10 active fighters based on Glicko rating. Active fighters are fighters who have fought within the last 365 days.

5.2 Changes to Fighter Skill Levels Over Time

When the UFC first began nearly 33 years ago, there were no MMA fighters per se, rather fighters came from other disciplines such as boxing or wrestling since the concept of mixing martial arts styles was still novel. This would limit a fighter’s theoretical combat toolbox as they would only be able to utilize their own martial arts-specific techniques and tactics. Even today, many UFC fighter’s fighting styles still rely heavily on their original martial arts and are reluctant to learn others to the detriment of their own careers. It is a relatively recent phenomenon for up-and-coming fighters to start their careers by training exclusively in MMA. Consequently, it is reasonable to believe that the skills of UFC fighters have improved and will continue to as more and more fighters move out of their elements and pursue pure MMA.

To prove this claim, we will conduct an ordinary least squares test on our fighter’s ratings—which roughly correlates to the skill of a fighter—over time. We first computed the average rating of every fighter who has fought in a specific

year for every single year the UFC has been around. Then, we set our threshold $\alpha = 0.05$ and our hypotheses as follows:

H_0 = UFC fighter ratings do not change over time (i.e., slope is 0)

H_a = UFC fighter ratings do change over time (i.e., slope is not 0)

Running the OLS test on our data results in a p-value of roughly 1.89×10^{-11} which is much smaller than α . Therefore, we reject H_0 and accept H_a as there is sufficient evidence that indicates that the ratings of UFC fighters are increasing over time.

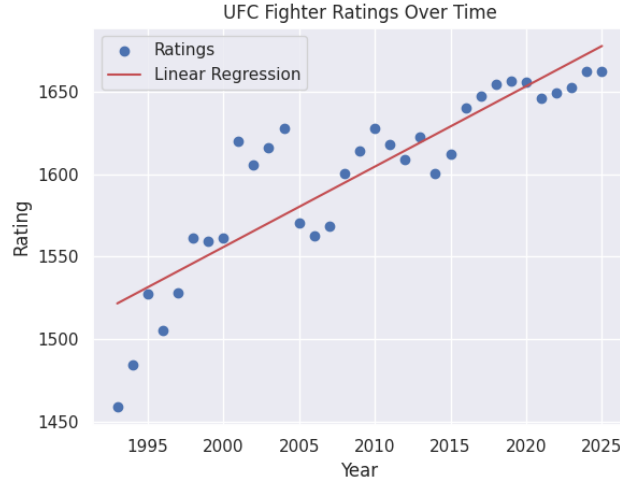


Figure 1: UFC fighters' average Glicko rating for every year since 1993.

The correlation coefficient r is approximately 0.88 which implies that there is a very strong positive relationship between the fighter's ratings and time. This can be seen in Fig.(1) as the linear regression line representing the average ratings of fighters clearly trends in a positive direction.

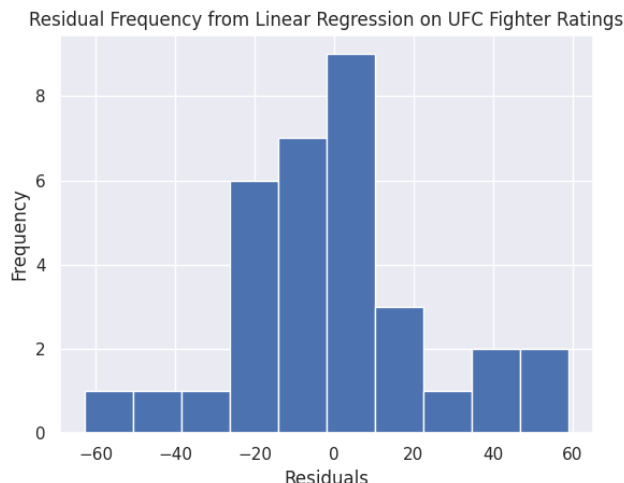


Figure 2: Histogram of residuals from applying linear regression to UFC fighter ratings and time.

As the OLS test requires that the residuals of linear regression have a normal distribution, we also conducted a normality test on them with $\alpha = 0.05$. The p-value returned from this test was about 0.42 which is larger than α . Thus, we can assume that the distribution of the residuals is normal. This can be seen in Fig.(2) where it looks roughly normally distributed since around two-thirds of the residuals are near the center and the rest are outside.

5.3 Advantages of Different Fighting Styles

In the following sections, §5.3.1 and §5.3.2, we will be exploring the advantages of different fighting styles and stances in the UFC to determine whether one is better than the other.

5.3.1 Southpaws Versus Orthodox Fighters

The conventional wisdom in the combat sports world tells us that orthodox fighters—fighters who throw jabs with their left hand and crosses with their right—struggle and often lose against southpaws—fighters who jab with their right and cross with their left. Southpaw stances are typically used by left-handed fighters as it allows them to throw crosses with their dominant hand. These fighters generally have several advantages over their orthodox counterparts. Assuming the proportion of left-handed people in the population of fighters follows that of the general population, only 10% of fighters would be southpaws which makes it difficult for orthodox fighters to find sparring partners or even have experience fighting southpaws, severely hindering their ability to prepare for a left-handed opponent. Contrasted with southpaws, who'll have no trouble finding orthodox

sparring partners and likely have been fighting right-handed fighters their entire life. Factors like these contribute to the popular opinion that southpaws will beat orthodox fighters the majority of the time.

However, these are just opinions. We will conduct a χ^2 test on UFC fighter stances and outcomes to substantiate whether this claim is true. We have also included switch hitters—fighters who switch between orthodox and southpaw stances—in our analysis. Firstly, we calculated the frequency of fight outcomes for each stance and stored them in a contingency table. Then, we set $\alpha = 0.05$ and defined our hypotheses as shown:

$H_0 =$ The proportions of UFC fights outcomes are the same for all stances.

$H_a =$ The proportions of UFC fights outcomes are not the same for all stances.

Conducting the χ^2 test on our contingency table, we receive the following p-value of 0.00751 which is smaller than α . Accordingly, we reject H_0 and accept H_a because we have significant evidence that shows us that a UFC fighter’s stance affects their fight outcomes.

Stance	Losses	Draws	Wins	Win Rate
Orthodox	6092	92	5892	48.7910%
Southpaw	1467	20	1625	52.2172%
Switch	426	6	468	52.0000%

Table 2: Contingency table of UFC fight outcomes with win rates for each fighting stance.

As it can be seen in Table 2, southpaw fighters win at much higher rates than their orthodox competitors. Switch hitters also perform better than orthodox fighters but not southpaws; this follows as training in both stances makes it much harder for a fighter to master the intricacies of both.

5.3.2 Strikers Versus Grapplers

In a reductive sense, every martial art can be categorized into two categories: striking-based—utilizing your hands, elbows, knees, and feet to strike an opponent, i.e. karate and boxing—and grappling-based—utilizing your body to manipulate and gain control of your opponent’s body, i.e., judo and Brazilian jiu-jitsu. A long-standing argument in the MMA community is whether striking or grappling is better. Currently, the community consensus is that grapplers are overwhelmingly dominating the UFC since seven out of the eight reigning UFC champions are predominantly grapplers.

We will determine whether there is evidence that confirms if this consensus is correct by performing another χ^2 test with $\alpha = 0.05$. However, since the UFC stats did not have data on what fighting style a fighter has, we will have to use our own heuristics: a grappler is a fighter that averages 0.3 takedowns and 30s of control time per round. This splits the fighters roughly in half with 1105

grapplers to 1479 strikers. Then, similarly to what we did in §5.3.1, we compute the frequency of each fighting style’s outcomes and store them in a contingency table. The hypotheses for the test are as detailed:

H_0 = The proportions of UFC fight outcomes are the same for all styles.

H_a = The proportions of UFC fights outcomes are not the same for all styles.

The p-value from the χ^2 test is approximately 1.96×10^{-26} which is clearly smaller than α . As a result, we reject H_0 and accept H_a because there is significant evidence that a fighting style affects the outcome of UFC fights.

Style	Losses	Draws	Wins	Win Rate
Grappler	3710	64	4400	53.8292%
Striker	4275	54	3593	45.3547%

Table 3: Contingency table of UFC fight outcomes with win rates for each fighting style.

From Table (3), it is evident that grapplers win much more frequently and at a higher rate than strikers. This essentially corroborates the sentiment that grappling-based fighters are completely dominating the UFC.

6 Predictor

6.1 Training a Model to Predict Fighter Matchups

From the fight data, ties are removed and unusable columns, such as red_rating, are dropped. Three classification models, SVM, Decision Tree and Random forest, were each trained a specified number of times. Their accuracy results were recorded and the best performing version of each model are saved as pickle files.

Model	Avg. Accuracy
SVM	0.899812
Forest	0.887915
Tree	0.844083

Table 4: Average accuracies of our models on the validation set over 40 runs.

6.2 Analyzing Predictor Results

To analyze the accuracies of our models, we will conduct an ANOVA test. First, we run a normality test on the predictor model accuracies and set $\alpha = 0.05$.

Model	P-value
SVM	0.44909419881106216
Tree	0.6693369215948202
Forest	0.04375017745202866

Table 5: Normality test on model accuracies.

As it can be seen in Table 5, the p-values of SVM and the decision tree both surpass α , allowing us to assume they are normal. The same cannot be said about the random forest model's p-value which barely misses the threshold.

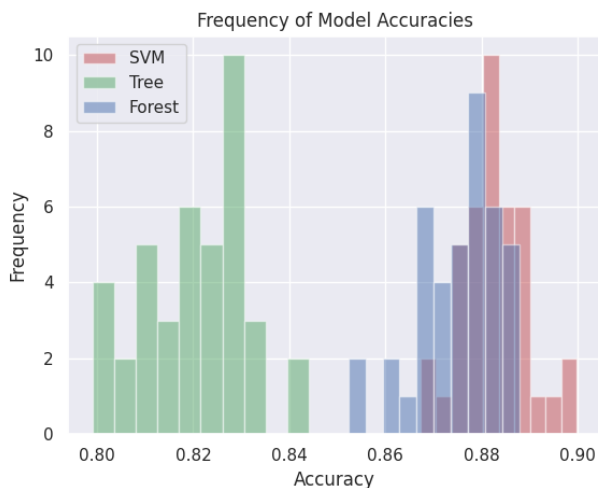


Figure 3: Histogram of our model accuracies over 40 runs.

From Fig.(3), the histogram of the random forest model looks somewhat normal and since there are 40 data points, we can just assume that it is normal and we can continue with our ANOVA test, setting our threshold $\alpha = 0.05$ and our hypotheses as follows:

$$H_0 = \text{The means of the model accuracies are equal}$$

$$H_a = \text{The means of the model accuracies are not equal}$$

The p-value obtained from ANOVA is roughly 3.42×10^{-61} which is way smaller than α . Therefore, we reject H_0 and accept H_a as there is significant evidence that the models do not have the same accuracy. We can also conclude that our SVM model produced the most accurate classifier as shown in Table 4.

Group 1	Group 2	Mean Diff.	P-adj	Lower	Upper	Reject
forest	svm	0.0073	0.0012	0.0025	0.0121	True
forest	tree	-0.055	0.0	-0.0598	-0.0502	True
svm	tree	-0.0623	0.0	-0.0671	-0.0575	True

Table 6: Tukey HSD post-hoc analysis on model accuracy.

6.3 Predictor Applications

In the following sections, we will be discussing two real-world applications of our predictor.

6.3.1 Predicting the Winners of UFC 314

Our matchup predictor takes the fighter data with a generated features file and a model. Then it prints the probability of each fighter winning. To test the correctness of our models in a real-world scenario, we used the SVM model to predict the winners of UFC 314, which was held on April 12th, 2025. Our models are trained from scraped data before the event began.

Red	Blue	Prediction	Actual
Alexander Volkanovski	Diego Lopes	Red (62.858%)	Red
Michael Chandler	Paddy Pimblett	Blue (80.140%)	Blue
Yair Rodriguez	Patricio Freire	N/A	Blue
Bryce Mitchell	Jean Silva	Blue (52.32%)	Blue
Nikita Krylov	Dominick Reyes	Red (64.648%)	Red
Dan Ige	Sean Woodson	Blue (94.990%)	Red
Yan Xiaonan	Virna Jandiroba	Red (57.478%)	Blue
Jim Miller	Chase Hooper	Blue (93.862%)	Blue
Darren Elkins	Julian Erosa	Red (62.457%)	Blue
Sedriques Dumas	Michal Oleksiejczuk	Blue (88.722%)	Blue
Sumudaerji	Mitch Raposo	Red (88.624%)	Red
TreSean Gore	Marco Tulio	Blue (83.861%)	Blue
Nora Cornolle	Hailey Cowan	Red (54.012%)	Red

Table 7: Predictions versus the actual results of UFC 314. Prediction column contains the winner and their probability of winning. Yair Rodriguez Vs. Patricio Freire could not be predicted as Freire is making his UFC debut and has no previous fights.

As shown in Table 7, our predictor managed to predict nine out of 12 or 75% of the UFC 314 fights correctly which is a shockingly good result.

6.3.2 Simulating a Men’s Lightweight Tournament

The UFC men’s lightweight division is currently one of the most stacked in the organization with several former champions and high-level contenders in the rankings. This should lead to many exciting matchups, but it seems like it is more of a catch-22; all of the top fighters want a shot at the champion and refuse to fight one another in fear of losing their standings in the lightweight division.

We aim to break this gridlock by using our predictor in a tournament simulator. Similar to the predictor, it takes the same arguments, with the exception of the input file where four or more fighters are expected, and an additional number of simulations to be run. The simulator first computes all possible matchup probabilities, then runs a Monte Carlo simulation to estimate the chance of each fighter reaching a certain milestone such as the quarterfinals and winning the entire tournament.

Fighter	QF	SF	Finals	Win
Ignacio Bahamondes	91.34%	83.26%	60.53%	51.64%
Dan Hooker	79.37%	48.74%	38.09%	11.07%
Dustin Poirier	86.48%	61.59%	25.24%	6.60%
Renato Moicano	90.10%	54.61%	18.42%	6.22%
Max Holloway	20.63%	12.28%	10.69%	5.81%

Table 8: Results of our UFC men’s lightweight tournament simulator after 10000 simulations. Only top 5 fighters are shown.

While we were testing our predictors with tournament simulations and untrained UFC events, we did run into some issues with suspicious outcomes, missed procedures like average scaling, and scattered undiscovered bugs. Additionally, we noticed that our Decision Tree model is unreliable for predicting UFC 314 event, most likely due to the large number of features. These are some improvements we would incorporate had we had more time and in retrospect.

7 Conclusion

In our report, we managed to provide clear answers to several burning questions from the MMA community. We first scraped data on UFC fights and fighters from their official stats website. Next, we sanitized our datasets and generated features for training our model as well as for analysis. A Glicko rating system was implemented to provide a quantifiable metric to rank fighters. We found that there was significant evidence that proved that the skill level of UFC fighters was increasing as time went on. Our experiments also showed that certain stylistic advantages, such as being a southpaw and or a grappler, affected the outcomes of their UFC matches to a statistically significant degree. Finally, we trained an SVM model to successfully predict the correct outcomes of 75% of the fights on the UFC 314 card.

8 Contributions

Joseph Choi:

- Used Discord to coordinate weekly meetings and monitor deadlines, ensuring the team remained aligned and on schedule for all major deliverables
- Utilized pandas to remove outliers, standardize fight formats, and convert control time and round lengths from strings to numerical values, producing a model-ready dataset with over 200 features
- Trained and cross-validated SVM, Decision Tree, and Random Forest classifiers using scikit-learn, comparing accuracy across 30+ runs to evaluate model consistency and effectiveness
- Applied grid search and manual tuning to identify optimal values (e.g., C for SVM, max_depth for trees), increasing model accuracy by up to 10%
- Reviewed code files to identify bugs, maintain modular design, and improve overall code readability and reliability

Jeffrey Jin:

- Assisted in the planning of the project to ensure our deliverables are completed on time
- Utilized BeautifulSoup to scrape 229 different features from all 8126 UFC fights as well as data for 4201 UFC fighters to provide thorough and accurate data to be used for training our model
- Implemented a custom Glicko rating system to provide quantifiable metrics for 2584 UFC fighters
- Summarized analysis and predictor results in Latex, producing an easily-accessible document for readers to enjoy

Clement Lau:

- Systematically cleaned over 4000 lines of fighter data by performing filtering and edge case handling, paving the way for feature generation
- Adjusted model parameters, training data, and model pipeline to create accurate prediction models with close to 90% accuracy
- Implemented the tournament and predictor scripts to run simulations with flexible input options, allowing unlimited hypothetical fights and tournaments

References

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